

RITVIK RAO

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EDUCATION

Ph.D. in Computer Science

2022 – present

University of Illinois Urbana-Champaign

Urbana, IL

- Advisor: Prof. Laxmikant V. Kalé; Thesis topic: optimizations for large-scale fine-grained irregular applications

B.S. in Computer Engineering and Computer Science

Dec. 2021

Northeastern University. Minor in Mathematics; GPA 3.91/4.0

Boston, MA

RESEARCH EXPERIENCE

Graduate Research Assistant

2022 – present

Parallel Programming Laboratory, University of Illinois Urbana-Champaign

Urbana, IL

- Primary designer of **Reconverse**, a scheduling and communication layer replacing Charm++'s 30-year-old Converse, with lottery-queue scheduling, queue registration, and an LCI machine layer; up to $2\times$ speedups on large-scale applications
- Developed ACIC, an asynchronous single-source shortest paths algorithm that uses continuous concurrent introspection and adaptation to reduce speculative work without constraining parallelism; up to $1.8\times$ faster than the state of the art
- Accelerated friends-of-friends cluster finding for cosmological simulations by combining tree-traversal and union-find libraries with fine-grained optimizations for highly imbalanced datasets
- Implemented the Ray Core API on top of **Charm4Py**, letting unmodified Ray programs use dynamic load balancing by mapping Ray actors to Charm4Py chares; $3\text{--}4\times$ speedups on Ray benchmarks over the standard Ray runtime

INDUSTRY AND LABORATORY EXPERIENCE

Ph.D. Extern

May 2025 – Aug. 2025

IBM Research

- Designed a portable collective communication library for AI workloads (all-gather, all-reduce, reduce-scatter), building on and contributing to the open-source UCCL library for moving large tensors between devices
- Combined UCCL with Charm++ so that collectives tolerate accelerator failures and adapt to any cluster's hardware and network topology; demonstrated on a distributed tensor multiplication written in Charm++

Ph.D. Intern

May 2023 – Aug. 2023

Pacific Northwest National Laboratory

- Developed a minimal-communication asynchronous algorithm for distributed reinforcement learning training on GPUs, built with RLlib on the Ray API, and applied it to speed up training for the influence maximization problem

Control Systems Engineering Co-op

Jan. 2021 – June 2021

Nuvera Fuel Cells

- Developed Simulink control software for a hydrogen fuel cell engine, raising power output from 45 to 60 kW and lowering the required starting temperature from 0°C to -30°C

GPU Software Engineering Co-op

Jan. 2020 – June 2020

Advanced Micro Devices (AMD)

- Designed a tool that instruments GPU shader programs during compilation to record performance metrics at run time

SKILLS

Languages: C++, C, Python, Java

Parallel and GPU programming: Charm++, Charm4Py, MPI, OpenMP, CUDA, ROCm/HIP, LCI, Ray/RLlib, UCCL

Tools: Slurm, Projections, Git, MATLAB/Simulink, $\text{\LaTeX}$

SELECTED PUBLICATIONS

- Joseph Schuchart, Patrick Diehl, et al. (21 authors, including **Ritvik Rao**). “A Survey of Distributed Asynchronous Many-Task Models and Their Applications.” *ACM Computing Surveys*, 2026.
- **Ritvik Rao**, Jiakun Yan, Aditya Bhosale, and Laxmikant V. Kalé. “Reconverse: A New Scheduling Library for Multithreaded Task-Based Runtime Systems.” In *PAW-ATM at SC26*, 2026. To appear.
- **Ritvik Rao** and Laxmikant V. Kalé. “Efficient Cosmological Friends-of-Friends Computation with Fine-Grained Optimizations.” In *IA³ at SC26*, 2026. To appear.
- **Ritvik Rao**, Kavitha Chandrasekar, and Laxmikant V. Kalé. “An Adaptive Asynchronous Approach for the Single-Source Shortest Paths Problem.” In *IA³ at SC24*, 2024.